

IP addressing and Subnetting

1. IP class and Range :

Class	IP Address Range	Purpose	Real-World Device/Service Example
Class A	1.0.0.0 – 126.255.255.255	Very large networks	Global ISPs or enterprise cloud data centers (e.g., AWS internal host 10.0.1.5)
Class B	128.0.0.0 – 191.255.255.255	Medium to large networks	University campus networks or corporate offices (e.g., main office router 172.16.1.1)
Class C	192.0.0.0 – 223.255.255.255	Small networks	Home Wi-Fi routers and connected smart devices (e.g., home PC 192.168.1.50)
Class D	224.0.0.0 – 239.255.255.255	Multicast traffic	Live video streaming / IPTV feeds sent to multiple subscribers at once (224.0.0.1)
Class E	240.0.0.0 – 255.255.255.255	Experimental / Research	Networking labs testing prototype protocols (e.g., host 240.1.2.3)

2. Network Calculations :

□ Given:

- **IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0 (/24)
- **Calculation Steps:**

1. Binary Conversion:

- IP Address: 11000000.10101000.00001010.00101101
- Subnet Mask: 11111111.11111111.11111111.00000000

2. Network Address Calculation: Perform a bitwise AND operation between the IP and Subnet Mask:

- 11000000.10101000.00001010.00000000 → **192.168.10.0**

3. Broadcast Address Calculation: Keep the network bits and turn all host bits (last octet) to 1:

- 11000000.10101000.00001010.11111111 → **192.168.10.255**

4. Valid Host Range Calculation: The range between the network address (+1) and the broadcast address (−1).

3. Food Delivery Startup Subnetting Plan :

To divide the 10.0.0.0/24 network into 4 equal subnets, borrow **2 bits** ($2^2 = 4$ subnets).

- **New Subnet Mask:** /26 (255.255.255.192)
- **Block Size:** $256 - 192 = 64$ addresses per subnet.

Subnet Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191

Subnet Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

4. College Wi-Fi Subnetting Scenario :

- **Given:**
 - **Original IP Block:** 172.16.0.0/16
 - **Requirement:** At least 20 subnets.
- **Calculation Steps:**
 1. **Calculate Borrowed Subnet Bits (n):**
 - Find the smallest power of 2 such that $2^n \geq 20$.
 - $2^4 = 16$ (Too small)
 - $2^5 = 32$ (Sufficient: gives 32 subnets, satisfying ≥ 20).
 - **Bits borrowed:** $n = 5$ bits.
 2. **Calculate New Prefix Length & Subnet Mask:**
 - New Prefix Length: 16 (original) + 5 (borrowed) = /21
 - Binary Mask: 11111111.11111111.11111000.00000000
 - Decimal Subnet Mask: 255.255.248.0
 3. **Calculate Hosts Available Per Subnet:**
 - Remaining host bits: $32 - 21 = 11$ bits.
 - Total addresses per subnet: $2^{11} = 2048$.

- Usable hosts: $2^{11} - 2 = 2046$ (subtracting Network and Broadcast addresses).
- **Final Answer:**
 - **New Subnet Mask:** 255.255.248.0 (or /21)
 - **Usable Hosts per Subnet:** 2,046

5. Troubleshooting Assignment :

Networking Homework Help: Why IP 192.168.1.300 Fails

1. Identifying the Error:

An IPv4 address consists of 4 numbers (octets) separated by dots. Each octet is made up of **8 bits**, meaning the minimum possible decimal value is **0** and the maximum possible decimal value is **255** ($2^8 - 1$).

The IP address 192.168.1.300 contains **300** in the last position. Because 300 is greater than 255, it is an invalid IP address, and network hardware will reject it.

2. What a Valid IP Address Should Look Like:

To fit into the 192.168.1.0/24 subnet:

- The first three octets must stay 192.168.1.
- The fourth octet must be a number between **1** and **254** (0 is reserved for the network address, and 255 is reserved for broadcasting).

Corrected Example:

A valid IP address for your friend's device would be **192.168.1.30**.